

1 camera view

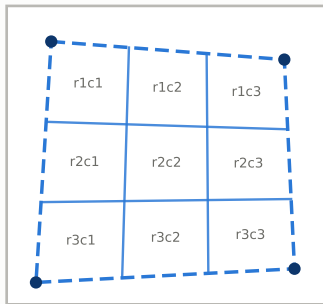
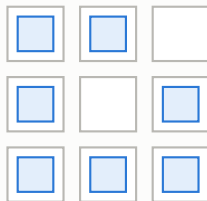


image space, no calibration:
operator clicks 4 corners, the grid
auto-splits into 3 × 3 cells
(adjustable, each rotatable)

2 crops and inference



each cell cropped upright with an 8 % margin,
letterboxed to 320 px, batched together

yolo26n detector, 2 classes
empty_box / filled_box

3 state and delivery

cell-state stabilizer
flip needs $\geq 70\%$ of a
15-vote window,
unknown after 5 s



one etagere_state message per rack
UDP + MQTT, topic {prefix}/etagere/{zone_id}, retained

consumers
warehouse management system (rack occupancy),
isicomms GET /etagere, operator dashboard matrix

the rack path never enters the metric floor pipeline: no homography, no triangulation, one camera per rack